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Crack propagation in the Double Cantilever Beam using Peridynamic theory

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ABSTRACT

In this study, Peridynamic (PD) theory is used to model mode-I delamination in unidirectional and multi-directional laminated composites. Experiments are conducted to determine mode-I fracture toughness in a unidirectional carbon-epoxy Double Cantilever Beam (DCB) specimen where the crack propagation remains on the original notch plane. The PD model of the DCB geometry is generated using an in-house pre-processor code in MATLAB and implemented in ABAQUS software. The brittle damage law in the original PD model is modified to a bilinear law to capture progressive softening. PD results are found to be in good agreement with the experimental results in terms of force–displacement curves and crack length. Next, this PD approach is applied to a multidirectional angle-ply DCB specimen. The PD model shows that delamination path jumps between the layers as the delamination grows. Force–displacement behaviour and delamination patterns obtained using PD model are compared with the corresponding experimental results from Gong et al. (2018). As a result, PD theory with bilinear softening law is found to successfully capture force–displacement relations and delamination migration in multidirectional laminated composites under mode-I loading conditions.

1. Introduction

Fibre-reinforced composites have a growing appeal in aerospace, automotive and energy industries due to their high strength-to-weight ratio. Physical properties of composite materials can be tailored according to requirements. Besides their advantage in lightweight design, composites exhibit complex failure mechanisms such as matrix cracks, fibre breakage and delamination. Delamination is a common failure mode in laminated composite structures which also interacts with other failure modes [1,2]. Composite engineering structures consist of multiple plies that are prone to delamination where delamination path may change due to coalescence of microcracks in the matrix. This behaviour is also known as delamination migration [3]. Gong et al. investigated delamination migration in multidirectional composites under Mode I static and fatigue loading [4]. Gong and his coworkers also studied delamination migration in multidirectional composites under Mode II loading [5]. They concluded that delamination migration under mode I loading is more stable than Mode II. For both Mode I and Mode II conditions, they found that ply splits develop inside the specimen which causes existing delaminations to find an ideal path with the lowest resistance [4,5].

Interface strength assessment is essential in order to investigate delamination propagation numerically. In Finite Element Analysis (FEA), special interface elements called cohesive elements are commonly used in order to analyse delamination propagation [6]. This special method is also known as Cohesive Zone Modelling (CZM) in the literature. Interface strength is associated with traction–separation law in CZM. The damage starts to develop when the traction between layers reaches a critical value. Li and Chen (2017), proposed an Extended Cohesive Damage Model (ECDM) to analyse multicrack failure behaviour in layered composites. Kumar et al. developed a Multi Phase-Field-Cohesive Zone formulation to investigate delamination migration [7]. Gong et al. developed an improved power law criterion to analyse large scale bridging in multidirectional laminates [8].

Peridynamic theory (PD) is also a robust numerical method to investigate failure in composites. It is a nonlocal continuum theory presented by Silling in 2000 [9]. PD equations of motion contain

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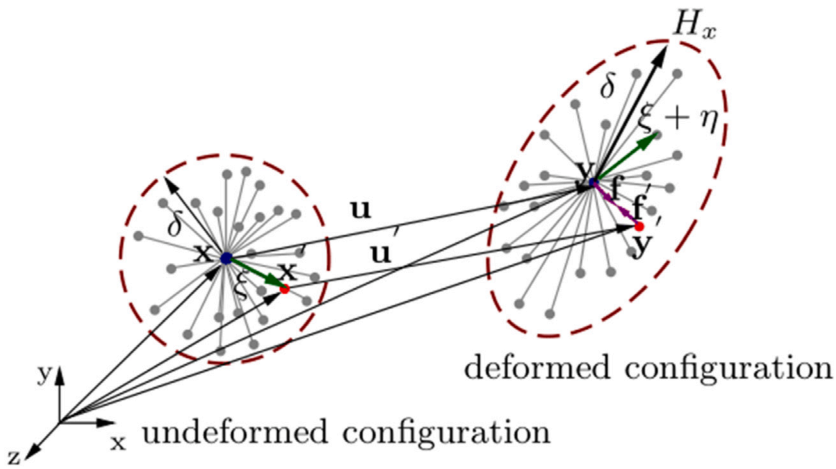


Fig. 1. Schematic representation of the BBPD model.

integral equations which are valid in the presence of discontinuities. Nonlocal interactions in a finite region called horizon, H, are considered in PD. The first PD model that employs pairwise interactions is known as bond-based PD (BBPD). The assumption of pairwise interactions leads to a fixed Poisson's ratio of $1/3$ for plane stress and $1/4$ for plane strain problems [10]. Later, Ordinary State Based PD (OSBPD) is formulated by Silling et al. which removes the constraint on the Poisson's ratio [11].

PD was successfully applied to various crack propagation problems in the literature. Silling and Askari (2005) introduced a micro elastic brittle damage model [12]. In their study, bond force was a linear function of bond stretch until the stretch between bonds reaches a critical value. If the bond stretch reaches the critical value, s_C, bond force sharply decreases to zero. Macek and Silling [13] proposed an approach to generate PD models using a finite element code ABAQUS with truss elements. Hu et al. determined the critical stretch of a bond relating to local deformation state which is also able to capture mode mixity [14]. Yolum et al. modified the original critical stretch law for brittle materials to a bilinear PD material law which accounts for bond softening [15]. Tong et al. investigated progressive failure in cohesive brittle materials using a PD model with exponential softening [16]. Yang et al. [17] developed a new OSBPD method based on CZM. They implemented the failure behaviour of the material using different softening curves. Their results have a good agreement with the experimental results and other numerical solution results.

The present paper aims to develop a practical PD simulation strategy to simulate delamination and delamination migration in composite laminates. Bilinear PD material is employed to account for the softening behaviour in the material interface. DCB experiments are conducted to validate the PD model. In Section 2, a brief review of the PD theory is given. Bilinear PD material model and implementation of PD in FEA software ABAQUS are also explained in Section 2. In Section 3, the methodology and the results of the DCB experiments to measure fracture toughness of unidirectional carbon-epoxy laminates are presented. In Section 4, the details and the results of the PD model for the tested DCB geometry are discussed and validated with the experimental results. In Section 5, the PD model with softening damage law is applied to a multidirectional DCB specimen where delamination is able to jump from the original notch plane to other interfaces. A PD grid space convergence study is presented for multidirectional DCB problem. In Section 5, comparison of total energies required for the crack propagation in PD simulations are compared with the results from the literature. The conclusion of this study is presented in Section 6.

2. Peridynamic (PD) theory

The equations of motion in Classical Continuum Mechanics (CCM) contains spatial derivatives that are undefined at discontinuities. Therefore, the original CCM formulation is not sufficient to consider imperfections such as cracks and voids. PD is a nonlocal version of Continuum Mechanics that uses integral terms instead of spatial derivatives in equations of motions as [9]

$$\rho(\mathbf{x})\,\ddot{\mathbf{u}}(\mathbf{x},t) = \int_H \mathbf{f}\left(\mathbf{u}' - \mathbf{u}, \mathbf{x}' - \mathbf{x}\right) dH + \mathbf{b}(\mathbf{x},t), \tag{1}$$

where ρ is the density, $\mathbf{u}$ is the displacement vector of material point $\mathbf{x}$, t is the time and $\mathbf{b}$ is the body force density. Note that, $\mathbf{f}\left(\mathbf{u}' - \mathbf{u}, \mathbf{x}' - \mathbf{x}\right)$ is the pairwise force density vector between material points $\mathbf{x}$ and $\mathbf{x}'$ as illustrated in Fig. 1.

In the BBPD theory, pairwise interactions between material points are called as PD bonds. Material point $\mathbf{x}$ interacts with other material points within a distance, δ, named as horizon, H_x as shown in Fig. 1. Force density vectors $\mathbf{f}$ and $\mathbf{f}'$ at material points $\mathbf{x}$ and $\mathbf{x}'$ are in opposite directions and they have equal magnitudes in order to assure conservation of angular momentum and linear momentum.

Relative position and relative deformation of material points are denoted with ξ and η, respectively in Fig. 1. They can be calculated as

$$\xi = \mathbf{x}' - \mathbf{x}, \tag{2}$$

$$\eta = \mathbf{u}(\mathbf{x}', \mathbf{t}) - \mathbf{u}(\mathbf{x}, \mathbf{t}). \tag{3}$$

In PD, energy stored in a single bond due to deformation is called micropotential, w [9]. The relation between the force density vector $\mathbf{f}(\eta,\xi)$, and micropotential, w, can be written for linear elastic materials as [12]

$$\mathbf{f}(\eta,\xi) = \frac{\partial w}{\partial \eta}. \tag{4}$$

Micropotential function for linear elastic materials is defined as

$$w(\eta,\xi) = \frac{cs^2|\xi|}{2}, \tag{5}$$

in which c denotes PD bond constant, s is the stretch between material points $\mathbf{x}$ and $\mathbf{x}'$ and can be calculated as [12]

$$s = \frac{|\eta+\xi| - |\xi|}{|\xi|}. \tag{6}$$

Using the relations in Eqs. (4) and (5), the force density vector, $\mathbf{f}(\eta,\xi)$ can be written as

$$\mathbf{f}(\eta,\xi) = \frac{\eta+\xi}{|\eta+\xi|}cs. \tag{7}$$

2.1. Constitutive model

The linear elastic behaviour of materials is characterized by the bond constant, c, in BBPD theory [12]. In linear elastic region, the bond interactions can be approximated as linear springs. The stiffness of PD bonds, c, can be calculated by equating the strain energy densities for PD theory and CCM [18]. For three dimensional models, the bond constant, c, can be calculated as [19]

$$c = \frac{18K}{\pi\delta^4}, \tag{8}$$

where δ is the radius of the horizon and K is the bulk Modulus of the material. K can be calculated as

$$K = \frac{E}{3(1-2\nu)}. \tag{9}$$

In the PD solution of DCB problem, Poisson's ratio is fixed at $1/3$ because of the plane stress assumption [10].

In this study, we model a narrow strip of the specimen across the width in which the crack propagates only along the length of the specimen. Position of the crack tip, the load–displacement and energy release rate calculations of composite materials depend on the elastic modulus of the layers in the fibre direction, E_{11}, as given in ASTM D5528 which is the standard test method for Mode-I fracture toughness of unidirectional composites [20]. Because of the aforementioned modelling of a narrow strip of the specimen in our PD model, we considered the layers to be isotropic for the purpose of calculating bond constants as obtained from Eq. (8). The elastic modulus, E, in Eq. (9) of each isotropic layer is set to E_{11}.

In order to consider failure in PD analysis, bond interactions are terminated if the stretch between material points, s, reaches a critical value s_c. At this point, the bond force is called as critical bond force, f_c, for the prototype microelastic brittle materials [12]. As originally suggested by [12], the critical stretch failure criteria represent brittle material failure as shown in Fig. 2(a). Later on, Silling et al. proposed a bilinear PD law that accounts for bond softening [21]. Zacariotto et al. employed bilinear PD law to analyse propagation of a pre-existing cracks in solids [22]. A bilinear PD law to simulate quasi-brittle fracture behaviour is also used in [23] where model parameters are calibrated to match the experimental results. Niazi et al. [24] calibrated the parameters of bilinear and trilinear PD laws to study crack nucleation in brittle solids. In this study, we employ bilinear PD law as shown in Fig. 2(a) since material interface have a softening behaviour under quasi static loading conditions. Note that bilinear material model is also commonly used in delamination simulations using CZM.

In the modified PD material model, the relation between the bond force, f, and stretch, s, is linear if the stretch, s, is smaller than s_0. For higher stretch values, bond failure starts with a linear softening behaviour and bond interactions are terminated completely when the stretch, s, reaches s_1 as shown in Fig. 2(a).

The micropotential at failure can be calculated using the area of the red triangle as

$$w = \frac{c s_0 s_1 \xi}{2}. \tag{10}$$

Energy release rate, G, required to break all PD bonds per unit area can be found by calculating the local strain energy density [12]. This can be done by integrating the micropotential, w, in Eq. (10) over a spherical coordinate system in a spherical cap as

$$G = \int_0^{\delta}\int_0^{2\pi}\int_z^{\delta}\int_0^{\cos^{-1}(z/\eta)} \frac{c s_0 s_1 \xi^3}{2} sin\phi d\phi d\xi d\theta dz. \tag{11}$$

where the angles ϕ, θ and the distance z as illustrated in Fig. 2(b). After evaluating the integral in Eq. (11), energy release rate can be found as

$$G = \frac{c s_0 s_1 \pi \delta^5}{10}. \tag{12}$$

The value of s_0 and s_1 should be determined in order to modify the bilinear law to the PD model. Energy release rate in Eq. (12) can be equated to the critical energy release rate, G_{IC} which is going to be found experimentally as will be described in Section 3.3. It is assumed that the ratio of Δ^f/Δ^0 from CZM application in [25] is the same as s_1/s_0 in the modified PD material model. This ratio is obtained as 577.67 by using the equations given in [25].

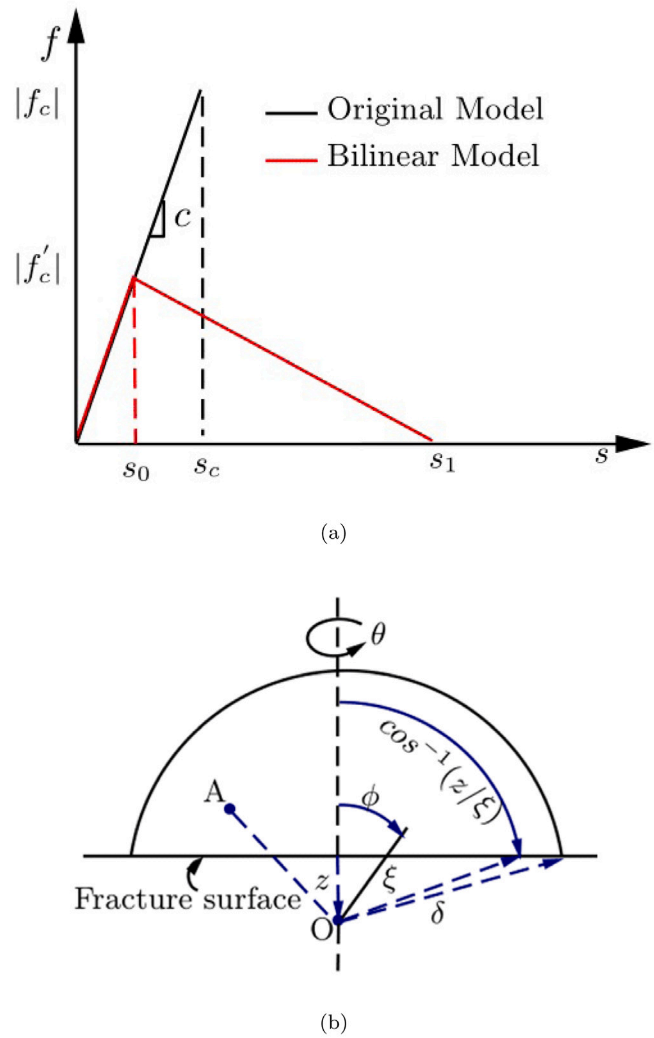


Fig. 2. Peridynamic material model: (a) Bilinear damage model, (b) evaluation of the fracture energy.

$$s_0 = \sqrt{\frac{10 G_{IC} \Delta^0}{\pi c \delta^5 \Delta^f}}. \tag{13}$$

After the value of s_0 is obtained, the failure behaviour of the material can be implemented for the PD model.

2.2. Implementation of PD theory in FEA

In BBPD, interactions between material points can be represented using linear springs or truss elements and mass properties of material points can be modelled with concentrated mass elements in commercial FEA softwares as suggested by Macek and Silling [13]. In this study, BBPD is implemented in ABAQUS software using T3D2 truss elements and CONM2 point mass elements. The PD model is generated using an inhouse MATLAB script. In the script, first geometry of the specimen is defined. The position and length of the notch is defined by the coordinates of start and end point of existing crack. Next, material points having grid spaces of Δx are generated. Then, family members

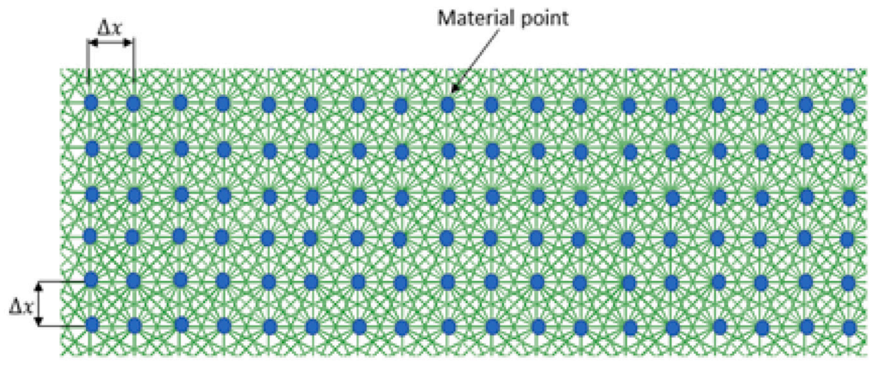


Fig. 3. Close-up view of PD bonds.

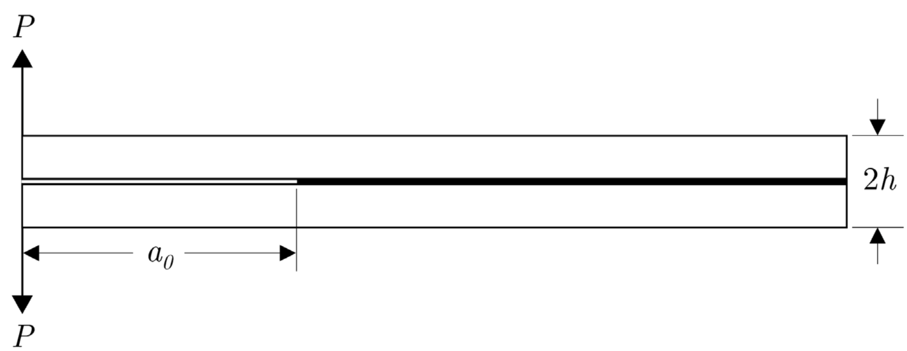


Fig. 4. DCB test configuration.

of each material point are identified according to the horizon size, δ. Finally, each material point is linked with its family members to obtain truss elements representing PD bonds. Close-up view of PD bonds is illustrated in Fig. 3 where Δx represents grid space between material points.

The stiffness of truss elements in Fig. 3 can be derived using the force equilibrium between truss elements and PD bonds as described in [13,26]. PD model of a structure with a bond constant, c, can be implemented in ABAQUS using the following properties:

$$E_t = c\Delta x^4, \qquad A_t = \Delta x^2, \tag{14}$$

where E_t is the elastic modulus and A_t is the cross sectional area.

3. DCB experiments

Double Cantilever Beam (DCB) tests are conducted in accordance with ASTM D5528 [20] to validate the PD predictions.

3.1. Material and specimen

Three DCB specimens are manufactured in the configuration shown in Fig. 4. The nominal thickness, $2h$, and width, a, values of the specimens are 2.4 mm and 25 mm, respectively. Each specimen is a unidirectional composite laminate consisting of 24 layers of 0° carbon fibre prepregs and a non-adhesive insert on a portion of the midplane which serves as a pre-crack of length a_0. The opening load, P, in Fig. 4, is applied to the specimen by means of piano hinges bonded on both surface of the DCB at the pre-cracked end.

The material used in manufacturing of DCB specimens is KOM13-A42-HSCF carbon fibre prepreg which is a product of KORDSA Composites for high strength engineering applications. The prepreg, which has 35% resin content, is a composition of OM13 thermoset resin system and a high strength unidirectional fabric consisting of DowAksa A42 carbon fibre tows each having 24,000 fibres. The elastic material properties of the carbon fibre prepreg are given in Table 1. The insert material on the midplane is a Kapton (polyimide) film with a thickness of 22 μm and provides a pre-crack length, a_0, of approximately 59 mm. After manufacturing of the specimens, more precise measurements of given quantities are performed using a Huvitz HDS-5800 digital microscope and these values are used in calculations.

Table 1
Material properties of KOM13-A42-HSCF unidirectional carbon fibre prepreg.

E_{11} [GPa]	E_{22} [GPa]	G_{12} [GPa]	ν_{12}
173	15	5.2	0.33

During the loading, the delamination length, a, should be measured on one side of the specimen. For easing the visual detection of the crack tip, both edges of DCB specimens are coated with a thin layer of white acrylic paint and a pattern of vertical lines 1 mm apart are created over the coated surface by spraying red acrylic paint through a laser-perforated ruler.

3.2. Experimental procedure

DCB tests are performed on a Shimadzu Autograph AGS-X 10 kN electromechanical axial testing machine in displacement control. It is noted that the load cell of this testing machine can measure the force with an accuracy of ±0.5% of the indicated force value over the range of 0.02 kN–10 kN. In the tests, DCB specimens are loaded at a constant crosshead speed of 1 mm/min and the force–displacement data were recorded at 20 Hz. Details of the test setup are shown in Fig. 5.

During the test, a Canon EOS 550D digital camera is positioned in front of the test setup by allowing it to observe in the same frame the instantaneous force and displacement values as well as the delamination onset. Digital zoom of the camera screen is brought to the maximum allowed level and the crack tip is followed by eye inspection.

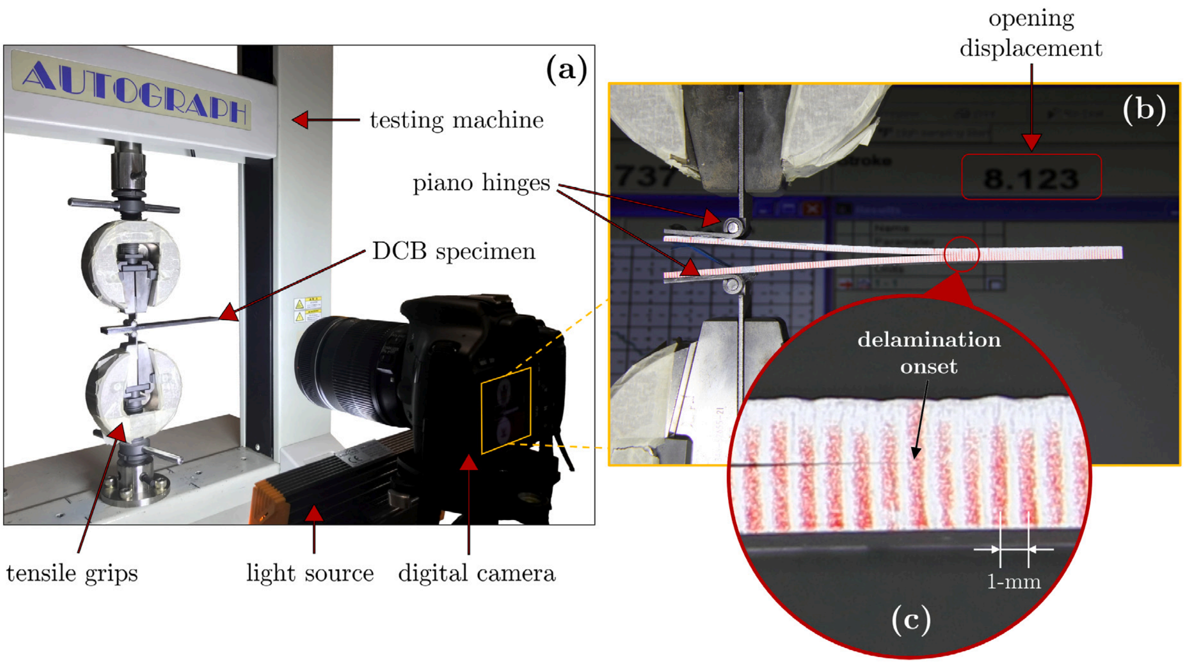


Fig. 5. Experimental setup for DCB tests: (a) testing equipment, (b) an image captured during the test, and (c) close-up view of the delamination onset region.

Table 2
Obtained critical energy release rates, G_{IC} [N/mm] values from DCB tests using different methods.

Specimen	Peak force [N]	G_{IC} [N/mm]		
		MBT	CC	MCC
DCB-1	87.15	0.280	0.303	0.305
DCB-2	81.15	0.321	0.323	0.324
DCB-3	87.00	0.295	0.337	0.337
Average	85.10	0.299	0.313	0.312
Standard deviation	3.42	0.021	0.010	0.010

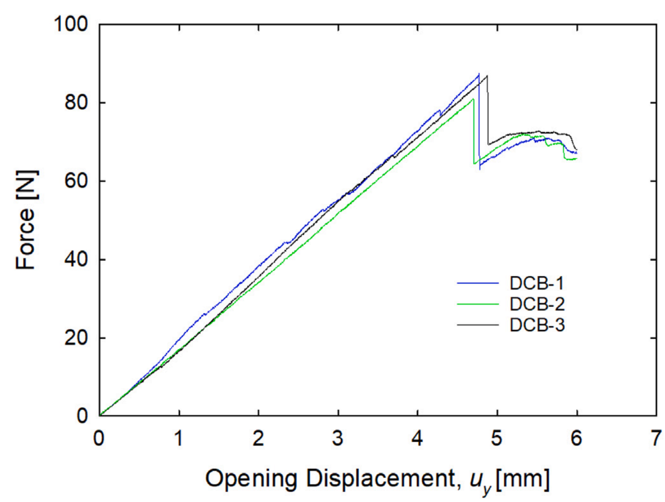


Fig. 6. Experimental force–displacement curves of $[0_{12}//0_{12}]$ DCB specimens.

Even though the in-situ real time pinpointing of exact location of the crack front is a challenging task, the use of a high-resolution digital camera enables capturing of high-resolution images (5184 × 3456 pixels) as delamination propagates every 1 mm distance. After the test, exact location of the crack front is measurable from these images within an accuracy of ±1 pixel via an on-screen pixel ruler such as Meazure v2.0.1. It is noted that, in our test setup configuration, 1 pixel corresponds to approximately 0.09 mm which, in fact, results in a better precision than the minimum suggested by ASTM D5528 [20].

3.3. Experimental results

Force–displacement curves obtained in three DCB tests are represented in Fig. 6. For all specimens, force–displacement curves have nearly identical slopes until a sharp load drop where delamination growth starts. Small local peaks which are observed on these constant slope regions indicate the detachment of the insert film from the pre-cracked interface. The ultimate peak forces in three tests are measured between 81–88 N with an average peak force of 85.10 N and a standard deviation of 3.42 N, as presented in Table 2. These two observations ensure the repeatability of the tests apart from the interface failure characteristics. The data beyond the sudden load drop are included in this plot until crack fronts propagate approximately 25 mm. In all tests, similar softening response can be observed following the initial delamination growth.

After the tests, Mode-I strain energy release rate, G_I, values were calculated using three data reduction methods suggested by the standard [20], namely the modified beam theory (MBT), the compliance calibration method (CC) and the modified compliance calibration method (MCC). It should be noted that G_I is evaluated for approximately initial 25 mm travel of the crack front, because at further distances, G_I values are likely to be affected by fibre bridging effects. Then, the Mode-I critical energy release rate, G_{IC}, of each specimen is determined as the average of the computed G_I values. G_{IC} values computed for each specimen with the three methods are presented in Table 2 with the average and standard deviation for each method. Average G_{IC} values determined by each method differs no more than %5 and the MBT method yields the most conservative G_{IC} values in accordance with the general observation [20].

4. PD analysis of DCB experiments

In this section, numerical simulations of DCB experiments presented in Section 3 are performed. Delamination propagation under static Mode I loading is investigated using BBPD.

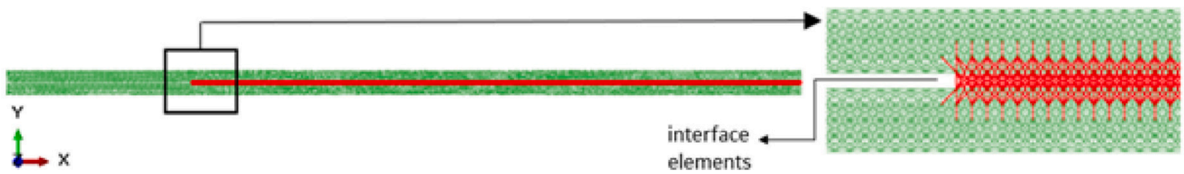


Fig. 7. PD truss elements created along the DCB specimen.

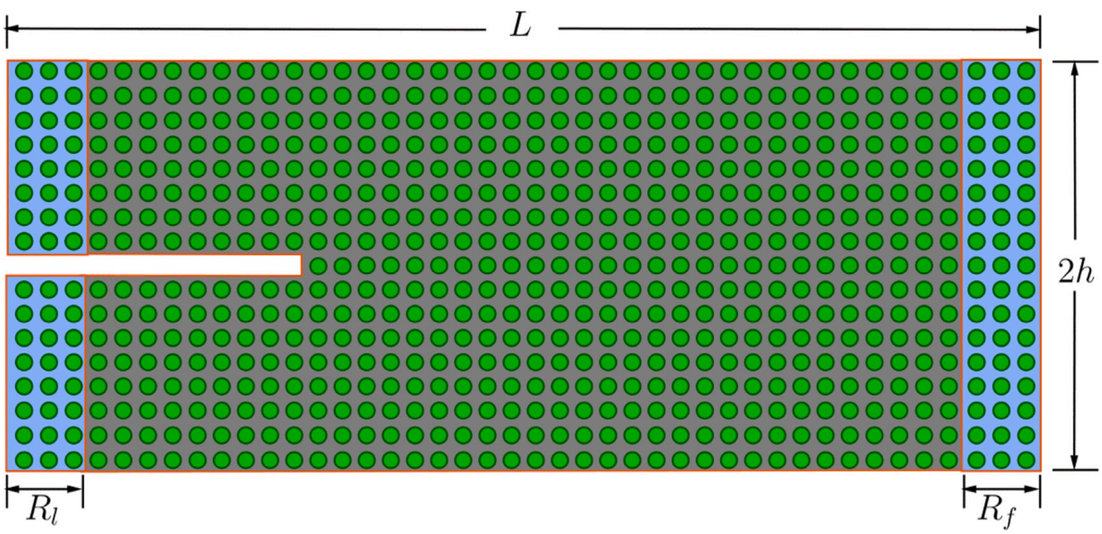


Fig. 8. Material points where boundary conditions are applied in the PD model (L = 170 mm, h = 2.5 mm).

4.1. PD model of the DCB test

PD model of the unidirectional composite DCB specimen is created as described in Section 2.2. The stacking sequence of the DCB test specimen is $[0_{12}//0_{12}]$ and the geometric properties are shown in Fig. 4. DCB specimen has a length of 170 mm and a width of 25 mm. Each arm has a thickness of 2.5 mm and the initial crack length is 59 mm. The material properties of the carbon/epoxy prepreg are given in Table 1.

The PD model of the DCB specimen shown in Fig. 7 is generated using an inhouse MATLAB pre-processing code. In the PD model, bonds are represented by T3D2 truss elements as suggested by Macek and Silling [13]. Plane stress assumption is used in order to reduce the computational cost. For this reason, a portion is modelled with only three material points across the width of the specimen. The effect of width is taken into account in the post processing.

The damage model with the bilinear softening behaviour in Fig. 2(a) is defined for the interface bonds shown with red in Fig. 7. The elements shown with green represent the upper and lower laminates. The value of G_{IC} is assumed to be 0.308 N/mm which is the average value of the G_{IC} values obtained from the three different calculation methods, as shown in Table 2. The boundary conditions are applied at the material points in the boundary regions, R_f and R_l, as shown in Fig. 8. In region R_f, the material points are restricted from all directions, and the opening displacements are applied in +y and −y directions at the top and bottom surface material points in region R_l, respectively.

In the PD analysis, nonlocal interactions are calculated within the horizon, H, with a radius of $\delta = m\Delta x$, where Δx is grid space between material points. The value of m is taken as 3.015 as suggested by [14]. The grid space, Δx is 0.5 mm for the PD analysis of DCB specimen.

4.2. PD results of the DCB test

In this section, results obtained from the experiments of the unidirectional DCB specimen and the corresponding PD analysis are compared in terms of force–displacement behaviours and delamination lengths. Force–displacement curve obtained from the PD analysis is presented in Fig. 9 in conjunction with three experimental curves. A good agreement can be observed regarding the initial slopes, peak forces, and post-damage responses contained in these curves.

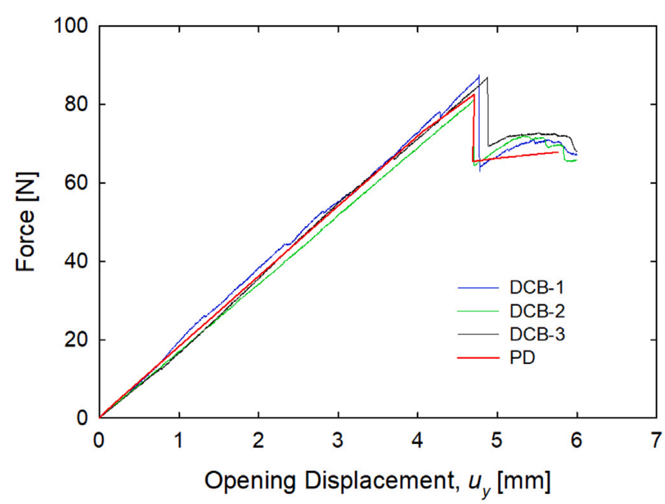
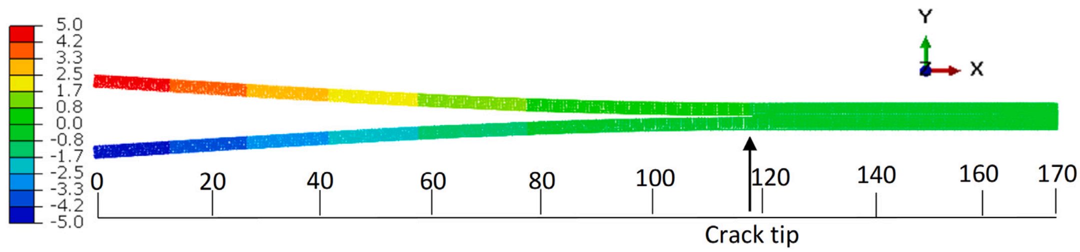
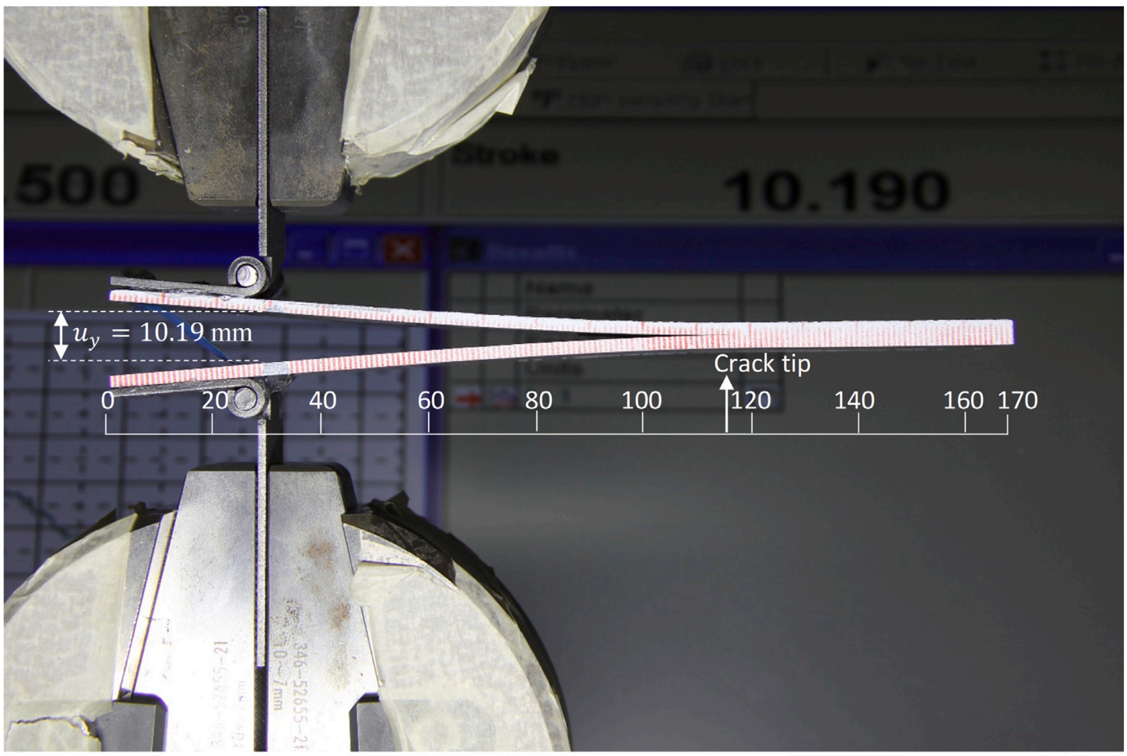


Fig. 9. Comparison of the force–displacement curves obtained from the PD analysis and DCB tests.

The measured peak forces in experiments were given in Table 2. In experiments, peak forces vary between 87.15 N and 81.15 N. The standard deviation and the average value of the experimental peak force are 3.42 N and 85.10 N, respectively. The peak force in the PD analysis is predicted as 82.29 N. It is noted that the PD analysis estimates the average experimental peak force within 3% error which indicates the agreement of PD results with experiments.



(a)



(b)

Fig. 10. Comparison of predicted and measured crack lengths at the same deformation state: (a) Displacement contours and the length scale showing the crack tip in the PD analysis, (b) crack tip position at the 10.19 mm opening displacement in the DCB test.

Fig. 10 presents a qualitative comparison of the experiments and PD analysis by means of delamination lengths measured at the same deformed state. In Fig. 10a and b, locations of the crack tips are marked with arrows and a length scale is located under the specimens. In the numerical analysis, the delamination length corresponding to 10 mm opening displacement is about 116 mm whereas the corresponding delamination length in the experiments is measured to be about 117 mm. Such agreement demonstrates that the PD model with the bilinear damage law is able to capture the Mode I delamination propagation in composite laminates successfully.

5. Multidirectional DCB specimen

In the previous section, delamination propagation along the original notched plane of a unidirectional DCB specimen was analysed. In this section, delamination propagation in multidirectional composite laminates under Mode I loading is investigated.

5.1. PD model for the multidirectional DCB specimen

The purpose of this investigation is to capture the delamination migration between the subsequent layers as the opening displacement in the multidirectional DCB specimen increases. The delamination migration phenomenon was previously observed in the experiments of Gong et al. [4]. Therefore, we analyse the multidirectional DCB configurations of Gong et al. [4]. The stacking sequence of the test specimen is given as

Table 3
Carbon epoxy material (IM7/8552) properties [4].

E_{11}	$E_{22} = E_{33}$	$G_{12} = G_{13}$	G_{23}	$\nu_{12} = \nu_{13}$	ν_{23}
161 GPa	11.38 GPa	5.17 GPa	3.98 GPa	0.320	0.436

* $G_{IC} = 0.2$ [N/mm], $G_{IIC} = 1.0$ [N/mm].

$$\left[\theta/-\theta/0/-\theta_4/-\alpha/-\theta/-\alpha/90/-\alpha/\alpha/90/\alpha/\theta/\alpha/-\theta_4/0/\theta/-\theta\right]_s$$

where θ is the orientation angle and $\alpha = 90 - \theta$. In this study, PD analyses are conducted for $\theta = 60°$ and $\theta = 75°$. The specimen has a length of 150 mm and a width of 20 mm. Initial crack length is 35 mm and each arm has a thickness of 3.5 mm. The material properties are given in Table 3 [4].

The multidirectional DCB problem is solved by coupling PD and FEA techniques in order to reduce the computational effort. Similar to the

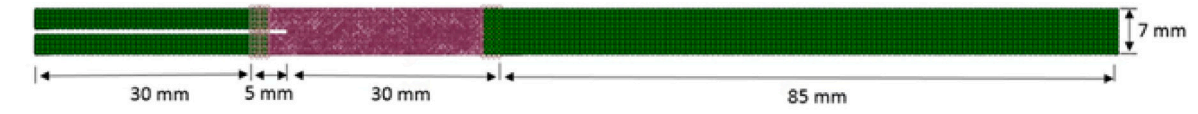


Fig. 11. Coupled FEA-PD model to analyse the multidirectional delamination problem.

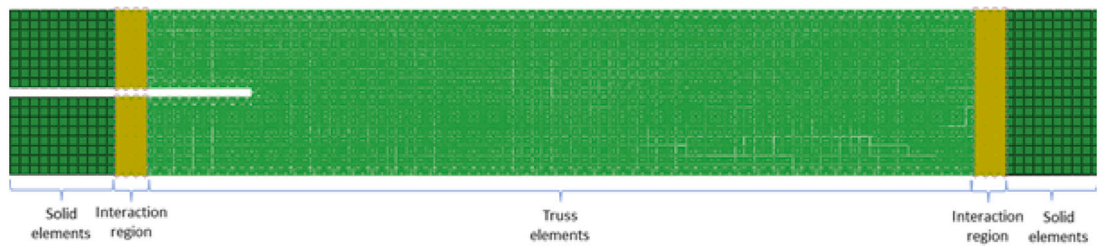


Fig. 12. Truss and solid elements and their interaction regions.

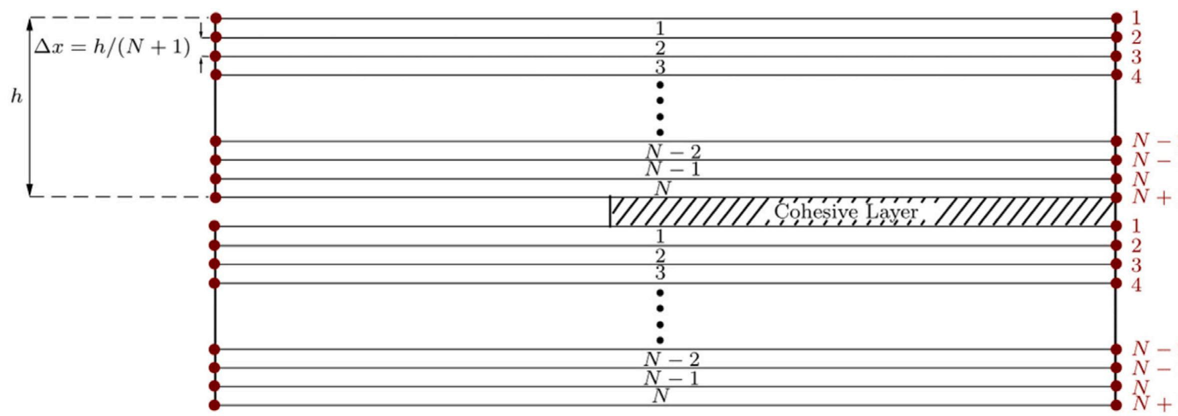


Fig. 13. Sublaminates of a multidirectional DCB specimen in the PD model.

unidirectional case, three material points are used across the width of the specimen, and then the results are scaled to the full width of the specimen during the post processing. Fig. 11 depicts the discretization of the multidirectional DCB specimen modelled with the coupled FEA-PD method. The first 30 mm of the specimen from the notched end is modelled using FEA with C3D8R solid elements in ABAQUS. The PD region starts 5 mm behind the initial crack tip and the length of the PD region is 35 mm. The rest of the specimen is modelled using FEA. The FEA and the PD regions are connected in the interaction region using kinematic couplings as shown in Fig. 12. The length of the interaction region is equal to the horizon size, $\delta = 3.015\Delta x$.

The multidirectional DCB specimen is divided into N sub-laminates as illustrated in Fig. 13. Therefore the corresponding grid space becomes $\Delta x = h/(N+1)$. The equivalent material constants of each sub-laminate are calculated by using the classical lamination theory. The extensional stiffness matrix, $[A]$, and extensional compliance matrix, $[A^*]$, are calculated as [27]

$$A_{ij} = \sum_{k=1}^{n} [\overline{\mathbf{Q}}_{ij}], \qquad i = 1,2,6; \quad j = 1,2,6, \tag{15}$$

$$[A^*] = [A]^{-1}, \tag{16}$$

where h_k is the thickness of the ply k and $\overline{\mathbf{Q}}_{ij}$ is the transformed reduced element stiffness matrix. The equivalent engineering constants of the laminate, namely, elastic modulus in longitudinal direction, E_x, elastic modulus in transverse direction, E_y, in-plane shear modulus, G_{xy}, and the in-plane Poisson's ratio, v_{xy}, can be calculated using the extensional compliance matrix defined in Eq. (16) as

$$E_x = \frac{1}{hA^*_{11}}, E_y = \frac{1}{hA^*_{22}}, G_{xy} = \frac{1}{hA^*_{66}}, v_{xy} = -\frac{A^*_{12}}{A^*_{11}}. \tag{17}$$

The equivalent engineering constants in Eq. (17) are substituted into Eq. (8) in order to calculate the bond constant c of the material. Then, the elastic modulus and cross-sectional area of the truss element are calculated using Eq. (14).

The bilinear damage model given by Eq. (13) is used for simulating the interface failure. The ratio s_1/s_0 is assumed to be the same as the ratio Δ^f/Δ^0 suggested by Turon et al. [25]. The critical energy release rate, G_{IC}, of the material is given as 0.2 N/mm in Ref. [4]. However, this is the nominal G_{IC} value which is measured at the interface of two unidirectional layers. It is worth to mention that the fracture toughness of multidirectional laminates are usually higher than that of a unidirectional laminate. On this issue, Naghipour et al. [28] and Rzeczkowski [29] reported that the value of G_{IC} is dependent on the stacking sequence of the laminate, and generally, G_{IC} of a multidirectional laminate is considerably higher than that of a unidirectional laminate. Experimental studies of Gong and his coworkers on the delamination in carbon/epoxy laminates with unidirectional and multidirectional interfaces also show that multidirectional laminates exhibit higher G_{IC} [30,31]. This phenomenon can be attributed to the degree of fibre bridging and crack surfaces created by matrix cracking in multidirectional laminates [32].

5.2. Parametric study of G_{IC}

To investigate the effect of the G_{IC} value on the Mode I delamination behaviour in multidirectional DCB specimens, we conducted parametric simulations for G_{IC} values ranging between 0.2 N/mm and 0.55 N/mm. In these simulations, a fixed grid space of $\Delta x = h/5$ is used. The resulting force–displacement curves obtained from the parametric

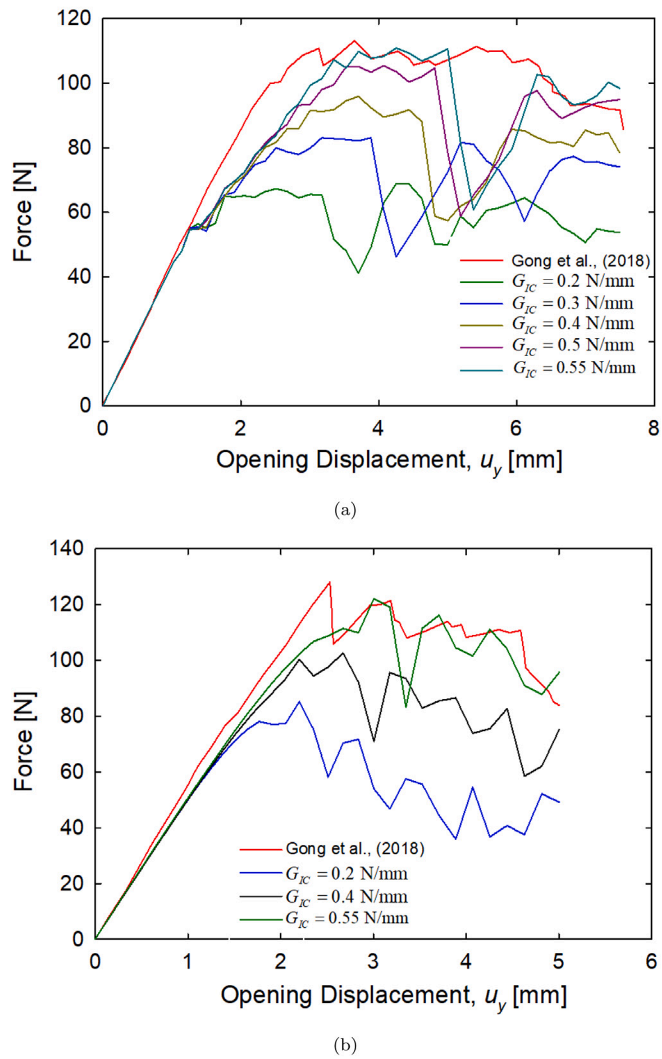


Fig. 14. Force–displacement curves obtained using different G_{IC} values: (a) $\theta = 60°$ and (b) $\theta = 75°$.

Table 4
The number of elements in the models created for the multilayer delamination problem.

(Δx)	PD bond interactions	Number of sub-laminates in PD model
$h/5$	233 104	6
$h/9$	1 590 482	3
$h/13$	4 853 552	2
$h/25$	17 923 278	1

PD analyses in conjunction with the experimental curve are presented for $\theta = 60°$ and $\theta = 75°$ in 14(a) and Fig. 14(b), respectively. For both orientation angles, the force–displacement curves obtained using $G_{IC} = 0.55$ N/mm represent closer behaviours to the experimental curves. Therefore, in the remaining analysis, the critical strain energy release rate of the multidirectional laminate is fixed as $G_{IC} = 0.55$ N/mm.

5.3. Grid space convergence study

Following an initial load drop where delamination starts to propagate, both experimental and numerical force–displacement curves represent a zig-zag behaviour, as seen in Figs. 14(a) and 14(b). It can be observed that the amplitudes of these fluctuations in numerical force–displacement curves are higher than those observed in the experimental

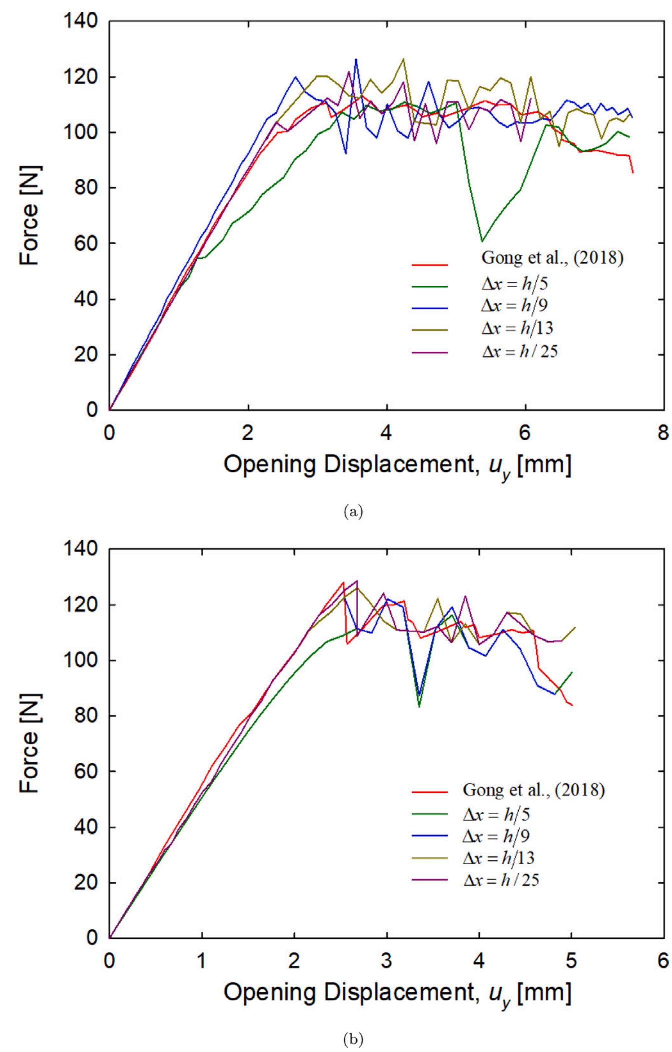


Fig. 15. Force–displacement curves from the grid space convergence study: (a) $\theta = 60°$ and (b) $\theta = 75°$ ($h = 3.5$ mm).

Table 5
External work using different G_{IC} values: $\theta = 60°$, $\Delta x = h/5$.

	External work [N.mm]
Gong et al. [4]	664.7
$G_{IC} = 0.2$ N/mm	402.3
$G_{IC} = 0.3$ N/mm	479.2
$G_{IC} = 0.4$ N/mm	529.6
$G_{IC} = 0.5$ N/mm	578.1
$G_{IC} = 0.55$ N/mm	600.8

curve. To investigate the effect of the grid space value on the amplitude of the fluctuations, we conducted a convergence study using different grid spaces of $\Delta x = h/5, h/9, h/13$ and $h/25$ while the horizon size is fixed as $\delta = 3.015\Delta x$ as suggested by the Ref. [14]. As the grid space, Δx, decreases, the number of bond interactions and sub-laminates in the PD model increases, as shown in Table 4. The analyses with the grid spaces $\Delta x = h/5$, $\Delta x = h/9$, $\Delta x = h/13$ and $\Delta x = h/25$ are completed in approximately 0.7 h, 10.5 h, 15 h and 23 h respectively using 6 CPU in HP Z620 workstation.

Force–displacement curves of the PD analyses with the different values of grid spaces are presented in Fig. 15. For $\Delta x = h/5$, it is observed that the elastic portions of the curves are partly consistent with the experimental load displacement curve. However, amplitudes

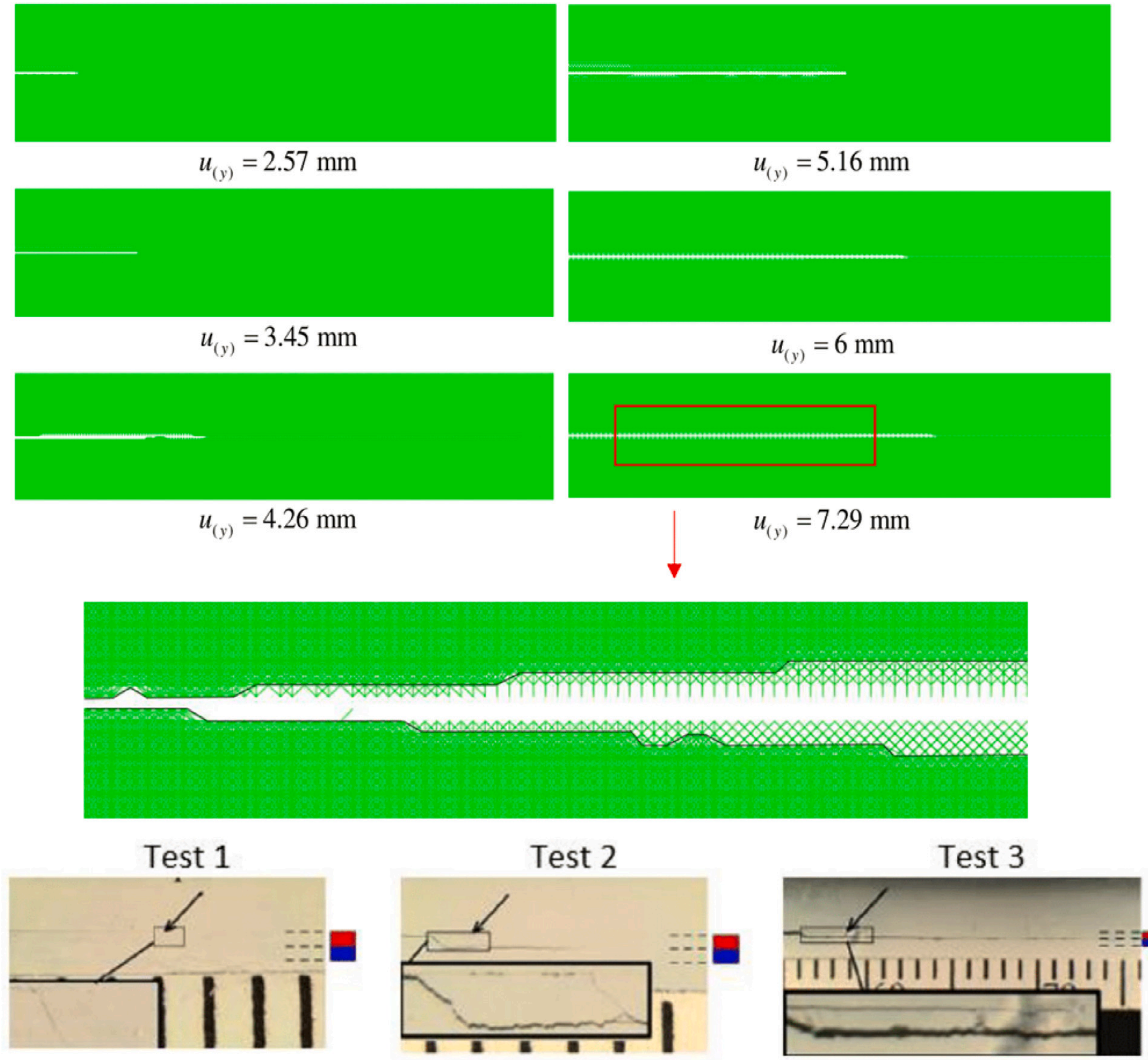


Fig. 16. Sequence of delamination propagation in the PD analysis of the multidirectional DCB specimen with $\theta = 60°$, close-up view of the final delamination pattern and the corresponding experimental images by Gong et al. [4] ($\Delta x = h/25$ and $h = 3.5$ mm).

of the zig-zag fluctuations in the damage propagation region are significantly greater with this grid space. The results obtained with grid spaces $\Delta x = h/13$ and $\Delta x = h/25$ are in better agreement with the experimental result of Gong et al. [4]. Convergence to the experimental force–displacement curve is achieved using $\Delta x = h/25$ where each layer in the DCB specimen is modelled separately.

5.4. External work

In this section, the external work done in the PD simulations is calculated at the maximum opening displacements of the experiments of Gong et al. [4]. Note that the external work needed for a specific opening displacement is equal to the area under the force–displacement curves. The external work for the force–displacement curves in Fig. 14 for $\theta = 60°$ and $\theta = 75°$ are calculated. The calculated energies for $\theta = 60°$ and $\theta = 75°$ are given in Tables 5 and 6 respectively. The external work in the experimental study of Gong et al. [4] was determined to be 664.7 N mm for $\theta = 60°$ and 439.1 N mm for $\theta = 75°$. It is seen that the closest values to these experimental results are obtained as 600.8 N mm using $G_{IC} = 0.55$ N/mm for $\theta = 60°$ with and error of 9.6% and 414.6 N mm using $G_{IC} = 0.5$ N/mm for $\theta = 75°$ with and error of 5.6%. These calculations were made for the grid size $\Delta x = h/5$.

Table 6
External work using different G_{IC} values: $\theta = 75°$.

	External work [N.mm]
Gong et al. [4]	439.1
$G_{IC} = 0.2$ N/mm	258.3
$G_{IC} = 0.4$ N/mm	350.2
$G_{IC} = 0.5$ N/mm	414.6

Table 7
External work using different critical energy release rates: $\theta = 60°$.

	External work [N.mm]	Error (%)
Gong et al. [4]	664.7	–
This study, $\Delta x = h/5$	600.8	9.6
This study, $\Delta x = h/9$	698.1	5.2
This study, $\Delta x = h/13$	694.5	4.9
This study, $\Delta x = h/25$	659.7	0.8

For finer grid sizes at a fixed $G_{IC} = 0.55$ N/mm for $\theta = 60°$ and $G_{IC} = 0.5$ N/mm for $\theta = 75°$, the external work values obtained are shown in Tables 7 and 8, respectively. It is seen in both tables that, the error percentage reduces as the grid space decreases. The lowest error

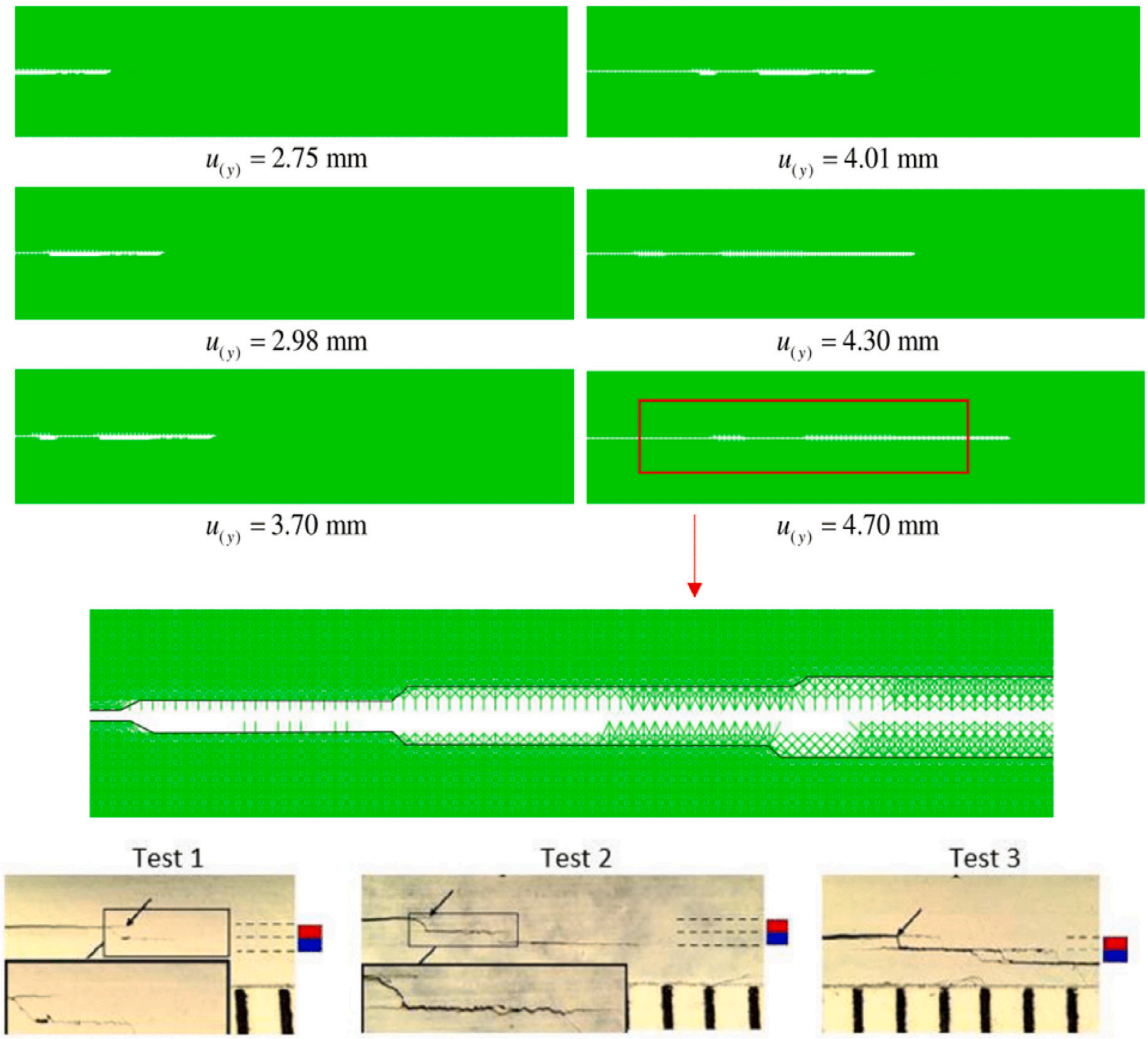


Fig. 17. Sequence of delamination propagation in the PD analysis of the multidirectional DCB specimen with $\theta = 75°$, close-up view of the final delamination pattern and the corresponding experimental images by Gong et al. [4] ($\Delta x = h/25$ and $h = 3.5$ mm).

Table 8
External work using different critical energy release rates: $\theta = 75°$.

	External work [N.mm]	Error (%)
Gong et al. [4]	439.1	–
This study, $\Delta x = h/5$	409.7	6.7
This study, $\Delta x = h/9$	414.6	5.6
This study, $\Delta x = h/13$	427.4	2.7
This study, $\Delta x = h/25$	430.7	1.9

is decreased to are 0.8 % for $\theta = 60°$ and 1.9 % for $\theta = 75°$ using the grid space $\Delta x = h/25$.

5.5. Investigation of crack patterns

Crack patterns in the multidirectional DCB specimens predicted by the PD analyses are compared with the patterns observed in the experimental images of Gong et al. [4]. In these analysis, the grid space of $\Delta x = h/25$ is used to capture a possible failure in each ply interface. Fig. 16 shows the sequence of the predicted crack growth in the multidirectional DCB specimen with $\theta = 60°$. In the close-up view of the damaged interface, it is observed that the delamination propagates on the original notched interface for a short distance and then migrates to the (−60°/60°) interface on the lower arm and (+60°/0°) interface on the upper arm. The PD prediction of the delamination pattern is found to be similar to the splitting patterns observed in the experimental images of Gong et al. [4] shown in Fig. 16.

The predicted sequence of delamination pattern in the multidirectional DCB specimen with $\theta = 75°$ is shown in 17. The close-up view of the damaged interface shows that the delamination propagates along the original notch plane until it splits into the lower −75° ply and reaches the (−75°/+75°) interface. After its propagation along this interface, it finally jumps to the (+75°/0°) interface passing through the lower +75° ply. Similar crack patterns are also observed in the experimental images of Gong et al. [4] for $\theta = 75°$.

6. Conclusion

In this study, delamination behaviour in unidirectional and multidirectional composite DCB specimen is investigated using BBPD. For the interface behaviour, a modified bilinear failure model is proposed. The obtained PD results are compared with the experiments conducted in RUUZGEM for unidirectional DCB and experiments from the literature for the multi-directional DCM specimen. The following conclusions are drawn:

- BBPD with modified bilinear material model is capable of capturing Mode I delamination behaviour in both unidirectional

and multi-directional laminated composites. In addition it is capable of delamination migration in multidirectional laminated composites.
- Based on external work done and crack pattern comparisons, it is observed that the best approximation to the experimental results can be obtained by using finer grid spaces. We suggest to model each layer separately in order to capture complex delamination patterns.
- The coupled FEA and PD provides a robust solution for sophisticated crack growth analysis.

CRediT authorship contribution statement

Ugur Yolum: Methodology, Analytics, Writing – original draft, Writing – review & editing. **Mirac Onur Bozkurt:** Methodology, Analytics, Writing – original draft, Writing – review & editing, Conducting experiments. **Eda Gok:** Methodology, Analytics, Writing – original draft, Writing – review & editing. **Demirkan Coker:** Conceptualization, Writing – review & editing, Supervision, Funding acquisition. **Mehmet Ali Güler:** Conceptualization, Writing – review & editing, Supervision, Funding acquisition.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

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Appendix. Uncertainity in the calculation of G_I

According to ASTM D5528 standard, the strain energy release rate of a perfectly built-in DCB specimen can be calculated as follows (we use the classical beam theory here for the sake of simplicity)

$$G_I = \frac{3Pd}{2ba}, \tag{A.1}$$

where P is the load, d is the load point displacement, b is the specimen width and a is the delamination length. For a sample input point from our date is given as:

$$P = 59.6 \pm 0.3 \quad \text{N}, \tag{A.2}$$

$$d = 6.42 \pm 0.02 \quad \text{mm}, \tag{A.3}$$

$$b = 25.10 \pm 0.01 \quad \text{mm}, \tag{A.4}$$

$$a = 71.83 \pm 0.01 \quad \text{mm}. \tag{A.5}$$

Among the values above, P and d have 3 significant figures while b and a have 4 significant figures. Therefore, G_I which is the multiplication/division of the parameters given above should have 3 significant figures. Therefore, strain energy release rate for this point can be written as

$$G_I = 0.318 \pm 0.03 \quad \text{N/mm}. \tag{A.6}$$

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